

Tyler Celotta

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Professional Summary

Physics student at Reed College enrolled in a dual degree program with a strong background in mechanical engineering and hands-on experience in machining and design. Skilled in Python programming, data analysis, and fabrication, with leadership experience, a love of the outdoors, and an engaged member of the community as an Eagle Scout.

Experience

Intern, Mechanical Engineering, Twin Cities Makers – Minneapolis, MN Jun 2025 – Aug 2025

- Designed and implemented a 15-hour curriculum covering manual milling, lathe turning, and CNC programming.
- Designed/built two-stage evaporative cooling prototype ($\Delta T = 5^{\circ}\text{F}$ at 60% humidity).
- Fabricated precision tools by hand (non-sparking propane regulator wrench, $\pm 0.003''$ tolerance).
- Conducted mechanical maintenance on shop systems (plumbing, HVAC, machines) to improve equipment uptime.
- Collaborated with shop members to brainstorm and troubleshoot project issues to help them move forward.

Engineering Job Shadow, Medtronic – Minneapolis, MN Oct 2023

- Shadowed 6 biomedical engineers developing pacemaker systems; observed live surgery for procedural analysis.
- Discussed testing and materials protocols for medical-grade components under NDA.

Grant Writer, Centennial School District – Lino Lakes, MN Nov 2021 – March 2022

- Co-drafted proposal for diesel-to-electric bus fleet conversion.
- Performed 30-year cost-benefit analysis with state data.
- Created technical and economic modeling in Excel to demonstrate long-term energy savings.

Lifeguard, City of Lino Lakes & Community Education ISD12 Jul 2022 – Aug 2023

Barista, Caribou Coffee Jun 2024 – Sept 2025

Van Driver, Reed College Aug 2024 – Present

Umpire, Centennial Little League Feb 2022 – Jul 2025

Portrait Photographer, Celotta Photos April 2023 – Present

Education

Reed College, BA in Physics – Portland, OR Jan 2024 – Present

- Enrolled in 3/2 Dual Degree Program: BA Physics (Reed) + BS Mechanical Engineering (Columbia University)
- Coursework: Calculus, Linear Algebra, Physics, Chemistry, Logic, Humanities, Ethics, Economics
- Earned credits from the University of Minnesota (Physics, English, Python)

Skills

Programming: Python (data analysis, math modeling, and graphical analysis)

Data Tools: R, Minitab, and Excel

Fabrication: Autodesk Fusion 360 (CAD/CAM), CNC milling (2.5 & 3-axis), manual milling (w/ rotary vice), MIG/TIG/Stick welding, equipment installation & calibration, small engine repair, and machine maintenance.

Certifications: OSHA 10, WFR, First Aid, CPR, Epinephrine (2025), Lifeguard (2022)

Achievements

- Eagle Scout, Boy Scouts of America (2023)
- Recognized for analytical clarity and team methodology in MathWorks Math Modeling (M3) Challenge (2023)
- Awarded SIA (summer internship award), which offers funding to support unpaid off-campus internship opportunities in the public sector and non-profits.

Interests

Community service (Boy Scouts of America, Eagle Scout), Photography, Mechanical Design and Fabrication, Public awareness campaigns (Arrive Alive Program & Save The Boundary Waters)